

Choose a big data storage technology in Azure

This article compares options for data storage for big data solutions—specifically, data storage for bulk data ingestion and batch processing, as opposed to [analytical datastores](#) or real-time streaming ingestion.

What are your options when choosing data storage in Azure?

There are several options for ingesting data into Azure, depending on your needs.

Unified logical data lake:

- [OneLake in Microsoft Fabric](#)

File storage:

- [Azure Storage blobs](#)
- [Azure Data Lake Storage Gen2](#)

NoSQL databases:

- [Azure Cosmos DB](#)
- [HBase on HDInsight](#) 

Analytical databases:

- [Azure Data Explorer](#)

OneLake in Fabric

[OneLake in Fabric](#) is a unified and logical data lake that's tailored for the entire organization. It serves as the central hub for all analytics data and is included with every Microsoft Fabric tenant. OneLake in Fabric is built on the foundation of Data Lake Storage Gen2.

OneLake in Fabric:

- Supports structured and unstructured file types.

- Stores all tabular data in Delta Parquet format.
- Provides a single data lake within tenant boundaries that's governed by default.
- Supports the creation of workspaces within a tenant so that an organization can distribute ownership and access policies.
- Supports the creation of various data items, such as lakehouses and warehouses, from which you can access data.

OneLake in Fabric serves as the common storage location for ingestion, transformation, real-time insights, and business intelligence visualizations. It centralizes various Fabric services and stores data items that all workloads use in Fabric. To choose the right datastore for your Fabric workloads, see [Fabric decision guide: choose a datastore](#).

Azure Storage blobs

Azure Storage is a managed storage service that is highly available, secure, durable, scalable, and redundant. Microsoft takes care of maintenance and handles critical problems for you. Azure Storage is the most ubiquitous storage solution Azure provides, due to the number of services and tools that can be used with it.

There are various Azure Storage services you can use to store data. The most flexible option for storing blobs from many data sources is [Blob storage](#). Blobs are basically files. They store pictures, documents, HTML files, virtual hard disks (VHDs), big data such as logs, database backups—pretty much anything. Blobs are stored in containers, which are similar to folders. A container provides a grouping of a set of blobs. A storage account can contain an unlimited number of containers, and a container can store an unlimited number of blobs.

Azure Storage is a good choice for big data and analytics solutions, because of its flexibility, high availability, and low cost. It provides hot, cool, and archive storage tiers for different use cases. For more information, see [Azure Blob Storage: Hot, cool, and archive storage tiers](#).

Azure Blob storage can be accessed from Hadoop (available through HDInsight). HDInsight can use a blob container in Azure Storage as the default file system for the cluster. Through a Hadoop Distributed File System (HDFS) interface provided by a WASB driver, the full set of components in HDInsight can operate directly on structured or unstructured data stored as blobs. Azure Blob storage can also be accessed via a Microsoft Fabric [Blob Storage shortcut](#).

Other features that make Azure Storage a good choice are:

- [Multiple concurrency strategies](#).
- [Disaster recovery and high-availability options](#).

- [Encryption at rest](#).
- [Azure role-based access control \(RBAC\)](#) to control access using Microsoft Entra users and groups.

Data Lake Storage Gen2

[Data Lake Storage Gen2](#) is a single, centralized repository where you can store all your data, both structured and unstructured. A data lake enables your organization to quickly and more easily store, access, and analyze a wide range of data in a single location. With a data lake, you don't need to conform your data to fit an existing structure. Instead, you can store your data in its raw or native format, usually as files or as binary large objects (blobs).

Data Lake Storage Gen2 converges the capabilities of Azure Data Lake Storage Gen1 with Azure Blob Storage. For example, Data Lake Storage Gen2 provides file system semantics, file-level security, and scale. Because these capabilities are built on Blob storage, you also get low-cost, tiered storage, with high availability/disaster recovery capabilities.

Data Lake Storage Gen2 makes Azure Storage the foundation for building enterprise data lakes on Azure. Designed from the start to service multiple petabytes of information while sustaining hundreds of gigabits of throughput, Data Lake Storage Gen2 allows you to easily manage massive amounts of data.

Azure Cosmos DB

[Azure Cosmos DB](#) is Microsoft's globally distributed multi-model database. Azure Cosmos DB guarantees single-digit-millisecond latencies at the 99th percentile anywhere in the world, provides multiple well-defined consistency models to fine-tune performance, and guarantees high availability with multi-homing capabilities.

Azure Cosmos DB is schema-agnostic. It automatically indexes all the data without requiring you to deal with schema and index management. It's also multi-model, natively supporting document, key-value, graph, and column-family data models.

Azure Cosmos DB features:

- [Geo-replication](#)
- [Elastic scaling of throughput and storage](#) worldwide
- [Five well-defined consistency levels](#)

HBase on HDInsight

[Apache HBase](#) is an open-source, NoSQL database that is built on Hadoop and modeled after Google BigTable. HBase provides random access and strong consistency for large amounts of unstructured and semi-structured data in a schemaless database organized by column families.

Data is stored in the rows of a table, and data within a row is grouped by column family. HBase is schemaless in the sense that you don't need to define the columns and the type of data stored in them before you use them. The open-source code scales linearly to handle petabytes of data on thousands of nodes. It can rely on data redundancy, batch processing, and other features that are provided by distributed applications in the Hadoop ecosystem.

The [HDInsight implementation](#) uses the scale-out architecture of HBase to provide automatic sharding of tables, strong consistency for reads and writes, and automatic failover. Performance is enhanced by in-memory caching for reads and high-throughput streaming for writes. In most cases, you want to [create the HBase cluster inside a virtual network](#) so other HDInsight clusters and applications can directly access the tables.

Azure Data Explorer

[Azure Data Explorer](#) is a fast and highly scalable data exploration service for log and telemetry data. It helps you handle the many data streams emitted by modern software so you can collect, store, and analyze data. Azure Data Explorer is ideal for analyzing large volumes of diverse data from any data source, such as websites, applications, IoT devices, and more. This data is used for diagnostics, monitoring, reporting, machine learning, and additional analytics capabilities. Azure Data Explorer makes it straightforward to ingest this data and enables you to do complex unplanned queries on the data in seconds.

Azure Data Explorer can be linearly [scaled out](#) for increasing ingestion and query processing throughput. An Azure Data Explorer cluster can be [deployed to a Virtual Network](#) for enabling private networks.

Key selection criteria

To narrow the choices, start by answering these questions:

- Do you need a unified data lake with multicloud support, robust governance, and integration with analytical tools? If yes, then choose OneLake in Fabric for simplified data management and enhanced collaboration.
- Do you need managed, high-speed, cloud-based storage for any type of text or binary data? If yes, then choose one of the file storage or analytics options.

- Do you need file storage that is optimized for parallel analytics workloads and high throughput/IOPS? If yes, then choose an option that is tuned to analytics workload performance.
- Do you need to store unstructured or semi-structured data in a schemaless database? If so, select one of the nonrelational or analytics options. Compare options for indexing and database models. Depending on the type of data you need to store, the primary database models might be the largest factor.
- Can you use the service in your region? Check the regional availability for each Azure service. For more information, see [Products available by region](#).

Capability matrix

The following tables summarize the key differences in capabilities.

OneLake in Fabric capabilities

 Expand table

Capability	OneLake in Fabric
Unified data lake	Provides a single, unified data lake for the entire organization, which eliminates data silos.
Multicloud support	Supports integration and compatibility with various cloud platforms.
Data governance	Includes features like data lineage, data protection, certification, and catalog integration.
Centralized data hub	Acts as a centralized hub for data discovery and management.
Analytical engine support	Compatible with multiple analytical engines. This compatibility enables diverse tools and technologies to operate on the same data.
Security and compliance	Ensures that sensitive data remains secure and access is restricted to authorized users only.
Ease of use	Provides a user-friendly design that's automatically available with every Fabric tenant and requires no setup.
Scalability	Capable of handling large volumes of data from various sources.

File storage capabilities

 Expand table

Capability	Data Lake Storage Gen2	Azure Blob Storage containers
Purpose	Optimized storage for big data analytics workloads	General purpose object store for a wide range of storage scenarios
Use cases	Batch, streaming analytics, and machine learning data such as log files, IoT data, click streams, large datasets	Any type of text or binary data, such as application back end, backup data, media storage for streaming, and general purpose data
Structure	Hierarchical file system	Object store with flat namespace
Authentication	Based on Microsoft Entra identities	Based on shared secrets Account Access Keys and Shared Access Signature Keys , and Azure RBAC
Authentication protocol	Open Authorization (OAuth) 2.0. Calls must contain a valid JWT (JSON web token) issued by Microsoft Entra ID	Hash-based Message Authentication Code (HMAC). Calls must contain a Base64-encoded SHA-256 hash over a part of the HTTP request.
Authorization	Portable Operating System Interface (POSIX) access control lists (ACLs). ACLs based on Microsoft Entra identities can be set file and folder level.	For account-level authorization use Account Access Keys . For account, container, or blob authorization use Shared Access Signature Keys .
Auditing	Available.	Available
Encryption at rest	Transparent, server side	Transparent, server side; Client-side encryption
Developer SDKs	.NET, Java, Python, Node.js	.NET, Java, Python, Node.js, C + +, Ruby
Analytics workload performance	Optimized performance for parallel analytics workloads, High Throughput and IOPS	Not optimized for analytics workloads
Size limits	No limits on account sizes, file sizes or number of files	Specific limits documented here
Geo-redundancy	Locally redundant (locally redundant storage (LRS)), globally redundant (geo-redundant storage (GRS)), read-access globally redundant (read-access geo-redundant storage (RA-GRS)), zone-redundant (zone-redundant storage (ZRS)).	Locally redundant (LRS), globally redundant (GRS), read-access globally redundant (RA-GRS), zone-redundant (ZRS). See Azure Storage redundancy for more information

NoSQL database capabilities

[Expand table](#)

Capability	Azure Cosmos DB	HBase on HDInsight
Primary database model	Document store, graph, key-value store, wide column store	Wide column store
Secondary indexes	Yes	No
SQL language support	Yes	Yes (using the Phoenix JDBC driver)
Consistency	Strong, bounded-staleness, session, consistent prefix, eventual	Strong
Native Azure Functions integration	Yes	No
Automatic global distribution	Yes	No HBase cluster replication can be configured across regions with eventual consistency
Pricing model	Elastically scalable request units (RUs) charged per-second as needed, elastically scalable storage	Per-minute pricing for HDInsight cluster (horizontal scaling of nodes), storage

Analytical database capabilities

[Expand table](#)

Capability	Azure Data Explorer
Primary database model	Relational (column store), telemetry, and time series store
SQL language support	Yes
Pricing model	Elastically scalable cluster instances
Authentication	Based on Microsoft Entra identities
Encryption at rest	Supported, customer-managed keys
Analytics workload performance	Optimized performance for parallel analytics workloads
Size limits	Linearly scalable

Contributors

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Next steps

- [What is Fabric](#)
- [Introduction to end-to-end analytics using Fabric](#)
- [Azure Cloud Storage Solutions and Services](#) 
- [Review your storage options](#)
- [Introduction to Azure Storage](#)
- [Introduction to Azure Data Explorer](#)

Related resources

- [Big data architectures](#)
- [Big data architecture style](#)
- [Understand data store models](#)

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